

Candidate Full Result: A Pathwise Karhunen–Loeve Expansion from the Separating Dual

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Status. **Candidate full result, likely valid, needs human review.** This packet gives a positive answer to both questions in Remark 2.26 of Cioica–Licht–Cox–Veraar [1]. The prior partial packet is now included only as provenance under

`provenance/partial_basis_from_dual_packet/`,

and proved only that the orthonormal basis can be chosen from E^* .

Source Question

The source paper proves a Karhunen–Loeve type expansion for Gaussian random variables in separable r -Banach spaces with separating dual, using Byczkowski’s RKHS theorem [2]. The relevant source passage is Remark 2.26 on page 14:

of [17, Theorem 1]. □

Remark 2.26. It is not immediately clear from [17] whether in the proof of Proposition 2.25 one may choose the orthonormal basis $(h_n)_{n \in \mathbb{N}}$ out of E^* . It is also unclear whether it is possible to choose $(h_n)_{n \in \mathbb{N}}$ in such a way that $\sum_{n=1}^{\infty} h_n(X) x_n = X$ $\mathbb{P}$ -a.s.

Now we explain how to use Proposition 2.25 prove the generalizations of Theo

In words, the source asks whether, in Proposition 2.25, the orthonormal basis (h_n) can be chosen from E^* , and whether it can be chosen so that

$$\sum_{n=1}^{\infty} h_n(X) x_n = X \quad \text{almost surely.}$$

Definitions and Source Tools

Let E be a separable r -Banach space, $0 < r \leq 1$, and let X be an E -valued Gaussian random variable with law μ . Following the notation recalled before Proposition 2.25 in the source paper, $E^2(\mu)$ denotes the real Hilbert space of square-integrable μ -a.e. additive measurable functionals on E , modulo equality μ -a.e., with inner product

$$\langle f, g \rangle_{E^2(\mu)} = \int_E f(x) g(x) d\mu(x).$$

Each $x^* \in E^*$ defines an element of $E^2(\mu)$. The proof uses the following source results: Proposition 2.25/Byczkowski's theorem for an in-law Gaussian series representation; the finite-matrix estimate and ideal property for Gaussian sums in r -Banach spaces; the theorem testing γ -radonifying operators against an orthonormal basis; and the Ito–Nisio convergence criterion, Proposition 2.18 in the source paper.

Proof Intuition

The first step is the density observation from the earlier partial packet: because E^* separates points and generates the Borel structure of E , the continuous linear functionals are dense in the Gaussian Hilbert space $E^2(\mu)$. Thus Gram–Schmidt gives an orthonormal basis $(h_n) \subseteq E^*$.

Byczkowski's theorem gives some Gaussian series

$$Y_0 = \sum_{j=1}^{\infty} \gamma_j y_j$$

whose limit has law μ . The coefficient sequence (y_j) may be viewed as a bounded Gaussian-radonifying map $R_0 : \ell^2 \rightarrow E$. For each dual-basis element h_n , the scalar series for $h_n(Y_0)$ has coefficient vector

$$a_n = (h_n(y_j))_{j \geq 1} \in \ell^2.$$

Since the $h_n(Y_0)$'s are independent standard Gaussians, the vectors (a_n) are orthonormal in ℓ^2 . Rotate the coefficient map by the isometry $e_n \mapsto a_n$ and set $x_n = R_0 a_n$. Then

$$h_k(x_n) = \delta_{kn}.$$

For every $x^* \in E^*$, the scalar coefficients $x^*(x_n)$ are precisely the Fourier coefficients of x^* in the basis (h_n) of $E^2(\mu)$. Therefore $x^*(\sum_{n \leq N} h_n(X)x_n) \rightarrow x^*(X)$ in L^2 for every $x^* \in E^*$, and Ito–Nisio turns this scalar convergence into the desired almost-sure E -valued convergence.

Lemma 1 (The coefficient operator). *Let $(y_j)_{j \geq 1} \subseteq E$. If the Gaussian series $\sum_{j=1}^{\infty} \gamma_j y_j$ converges almost surely to an E -valued Gaussian random variable, then the map $R_0 e_j = y_j$, initially defined on the finitely supported vectors in ℓ^2 , extends to a bounded operator $R_0 : \ell^2 \rightarrow E$. Moreover $R_0 \in \gamma(\ell^2, E)$.*

Proof. Let $S_N = \sum_{j=1}^N \gamma_j y_j$. By the source paper's Hoffmann–Jorgensen theorem and Fernique integrability for Gaussian random variables, $S_N \rightarrow S$ in $L^p(\Omega; E)$ for every $0 < p < \infty$; in particular, $\sup_N \|S_N\|_{L^1(\Omega; E)} < \infty$.

For a finitely supported vector $a = (a_1, \dots, a_N) \in \ell_N^2$ with $\|a\|_2 \leq 1$, apply the source paper's finite-matrix estimate for Gaussian sums to the $1 \times N$ matrix with row a . It gives

$$\left\| \gamma_1 \sum_{j=1}^N a_j y_j \right\|_{L^1(\Omega; E)} \leq C_r \left\| \sum_{j=1}^N \gamma_j y_j \right\|_{L^1(\Omega; E)} \leq C_r \sup_M \|S_M\|_{L^1(\Omega; E)}.$$

Since the left-hand side is $\mathbb{E}|\gamma_1| \|\sum_{j=1}^N a_j y_j\|_E$, the finite coefficient map is uniformly bounded on the unit ball of ℓ^2 . Hence it extends to a bounded operator $R_0 : \ell^2 \rightarrow E$. Finally, the testing-against-an-ONB theorem from the source paper applies to the standard basis of ℓ^2 ; because $\sum_j \gamma_j R_0 e_j = \sum_j \gamma_j y_j$ converges in L^p , it follows that $R_0 \in \gamma(\ell^2, E)$. $\square$

Theorem 1. *Let E be a separable r -Banach space whose dual E^* separates the points of E . Let X be an E -valued Gaussian random variable with law μ . Then there exist an orthonormal basis $(h_n)_{n \geq 1}$ of $E^2(\mu)$, with every $h_n \in E^*$, and a sequence $(x_n)_{n \geq 1} \subseteq E$, such that*

$$\sum_{n=1}^{\infty} h_n(X) x_n = X \quad \mathbb{P}\text{-almost surely.}$$

Proof. First, the canonical image of E^* is dense in $E^2(\mu)$. Indeed, let D be its closed span and take $g \in E^2(\mu)$ orthogonal to D . For any finite family $x_1^*, \dots, x_m^* \in E^*$, the random vector

$$(g(X), x_1^*(X), \dots, x_m^*(X))$$

is jointly Gaussian, and the first coordinate has zero covariance with all other coordinates. Hence $g(X)$ is independent of every finite dual coordinate vector. Since E is separable and E^* separates points, the source paper recalls that $\mathcal{B}(E) = \sigma(x^* : x^* \in E^*)$ and that a countable subset of E^* separates points. Thus $g(X)$ is measurable with respect to $\sigma\{x^*(X) : x^* \in E^*\}$, while the preceding finite independence implies independence from this sigma-field. Therefore $g(X)$ is independent of itself and hence is almost surely constant. As an element of $E^2(\mu)$, it is a centered Gaussian functional, so $g = 0$. Thus $D^\perp = \{0\}$, and $D = E^2(\mu)$. Applying Gram–Schmidt to a countable dense subset of the canonical image of E^* gives an orthonormal basis $(h_n) \subseteq E^*$. The finite-dimensional cases are identical with finite sums, so assume for notational simplicity that the basis is infinite.

By Proposition 2.25 of the source paper, choose a Gaussian sequence $(\gamma_j)_{j \geq 1}$ and vectors $y_j \in E$ such that

$$Y_0 := \sum_{j=1}^{\infty} \gamma_j y_j$$

converges almost surely and Y_0 has the same distribution as X . By the lemma, the coefficient map $R_0 : \ell^2 \rightarrow E$, $R_0 e_j = y_j$, is bounded.

For each n , continuity of h_n gives

$$h_n(Y_0) = \sum_{j=1}^{\infty} \gamma_j h_n(y_j) \quad \text{in } L^2(\Omega).$$

Set $a_n = (h_n(y_j))_{j \geq 1} \in \ell^2$. Since Y_0 and X have the same law and (h_n) is orthonormal in $E^2(\mu)$,

$$\langle a_n, a_m \rangle_{\ell^2} = \mathbb{E}[h_n(Y_0) h_m(Y_0)] = \mathbb{E}[h_n(X) h_m(X)] = \delta_{nm}.$$

Thus the map $J : \ell^2 \rightarrow \ell^2$ defined by $J e_n = a_n$ is an isometry. Define

$$x_n := R_0 J e_n = R_0 a_n, \quad n \geq 1.$$

For any k and any finitely supported $v \in \ell^2$,

$$h_k(R_0 v) = \langle v, a_k \rangle_{\ell^2}.$$

By continuity this remains true for all $v \in \ell^2$, and therefore

$$h_k(x_n) = \langle a_n, a_k \rangle_{\ell^2} = \delta_{kn}.$$

Now fix $x^* \in E^*$. Put $b = (x^*(y_j))_{j \geq 1} \in \ell^2$; this is the coefficient vector of the scalar L^2 -series for $x^*(Y_0)$. Then

$$x^*(x_n) = x^*(R_0 a_n) = \langle a_n, b \rangle_{\ell^2} = \mathbb{E}[h_n(Y_0)x^*(Y_0)] = \mathbb{E}[h_n(X)x^*(X)].$$

In other words, $x^*(x_n)$ is exactly the n -th Fourier coefficient of the element $x^* \in E^2(\mu)$ in the orthonormal basis (h_n) . Therefore the partial sums

$$S_N := \sum_{n=1}^N h_n(X)x_n$$

satisfy, for every $x^* \in E^*$,

$$x^*(S_N) = \sum_{n=1}^N h_n(X)x^*(x_n) \longrightarrow x^*(X) \quad \text{in } L^2(\Omega).$$

The random variables $(h_n(X))_{n \geq 1}$ form a Gaussian sequence, so the increments $h_n(X)x_n$ are independent and symmetric. The Ito–Nisio proposition in the source paper applies because E^* separates points. The scalar convergence just proved is precisely its dual weak convergence condition with candidate limit X . Hence $S_N \rightarrow X$ almost surely, which is the asserted pathwise expansion. $\square$

Verification Notes

This is a full candidate solution of Remark 2.26 under the same hypotheses as Proposition 2.25. The proof deliberately combines source-paper tools rather than presenting a later literature answer: the density argument gives the dual-valued ONB, Byczkowski’s in-law series gives initial coefficients, the finite-matrix estimate legitimizes the coefficient operator, and Ito–Nisio identifies the rotated series with the original X .

A bounded literature and duplication check was performed against the run indexes and exact web searches for the remark’s wording, “Karhunen–Loeve”, “Byczkowski”, “orthonormal basis out of E^* ”, and “ $h_n(X)x_n = X$ ”. No prior full packet or direct later answer was found.

Limitations and Review Recommendation

Human review should focus on two points. First, verify that the coefficient operator lemma is an acceptable extraction from the source paper’s finite matrix estimate, Hoffmann–Jorgensen theorem, Fernique theorem, and testing-against-an-ONB theorem. Second, verify the final Ito–Nisio invocation: the scalar L^2 convergence for every $x^* \in E^*$ should indeed imply almost-sure convergence of the E -valued partial sums to the specified random variable X .

References

- [1] P. A. Cioica-Licht, S. G. Cox, and M. C. Veraar, *Stochastic integration in quasi-Banach spaces*, arXiv:1804.08947.
- [2] T. Byczkowski, *RKHS for Gaussian measures on metric vector spaces*, Bull. Polish Acad. Sci. Math. 35 (1–2), 93–103, 1987.
- [3] T. Byczkowski and T. Inglot, *Gaussian random series on metric vector spaces*, Math. Z. 196 (1), 39–50, 1987.